1Sagay

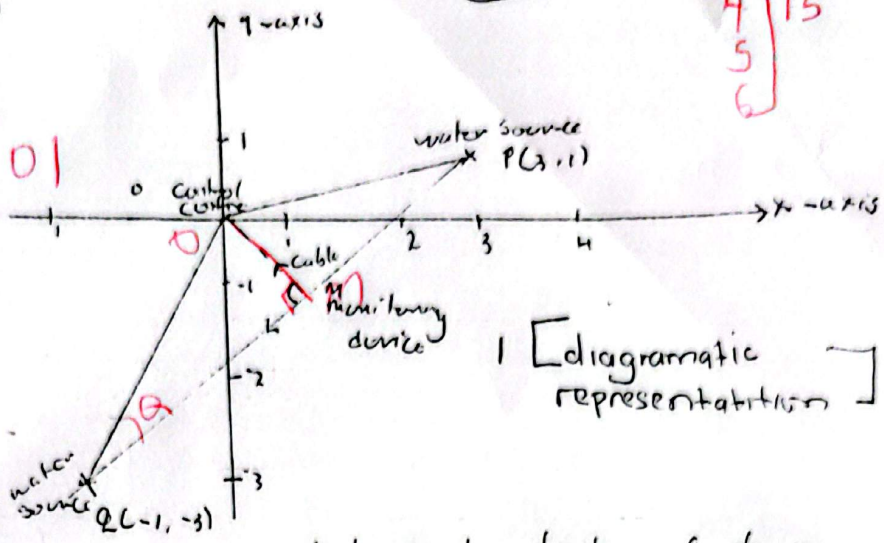
SUB MATHS

SCORING GUIDE

$\frac{x}{70}$

Item 1] 10
2
3] 15
4
5
6

Item 1
Sketch



any sketch T-01

1 [diagrammatic representation]

Let M be location of device

Co-ordinates of location of monitoring device; M m-01

$\left(\frac{3+-1}{2}, \frac{1+-3}{2}\right)$ 1 [Identifies mid-point]

$= (1, -1)$ 1 [Output] m-01 output

$\overrightarrow{MQ} = \begin{pmatrix} -1 \\ -3 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \overrightarrow{MQ} \begin{pmatrix} -2 \\ -2 \end{pmatrix}$ 1 [Identifies position vector] m-01

$\overrightarrow{MO} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ m-01

Angle $O\hat{M}Q$

$\overrightarrow{MQ} \cdot \overrightarrow{MO} = |\overrightarrow{MQ}||\overrightarrow{MO}| \cos\theta$ m-01 1 [Identifies dot product]

$|\overrightarrow{MO}| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$ units

$|\overrightarrow{MQ}| = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8}$ units } m-01

1 [Any Identifies Magnitude]

1 [Any correct magnitude]

$\cos\theta = \frac{\begin{pmatrix} -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ +1 \end{pmatrix}}{\sqrt{2} \cdot \sqrt{8}} = \frac{0}{\sqrt{16}} = 0$ m-01 1 [dot Out product]

$\theta = 90°$ m-01 1 [Correct output]

The signal transmission will be inefficient since the angle is more than acute. 1 [Decision]

TT : 10 scores D-01

Item 2

$R(t) = 12t - 3t^2$ ↑-01

$R'(t) = 12 - 6t$ ↗ 1 [Identifies differentiation]

For maximum or minimum, $R'(t) = 0$ 1 [For $R'(t)$ ...]

$12 - 6t = 0$ 1 [For $R'(t) = 0$]

$12 = 6t$ ↗ m-01

$t = 2$ hours 1 output for time.

$R''(t) = -6 < 0$ ∴ maximum 1 [For $R''(t)$]

↳ m-01 1 [Nature]

max Rate is when $t = 2$ hours

$R(2) = 12(2) - 3(2)^2$ ↗ m-01

$= 12$ Litres per hours 1 [correct output]

D-01 Since the maximum rate exceeds 10 litres per hour, irrigation will be successful 1 [correct Decision]

Let the amount of water remaining be Θ

$\frac{d\Theta}{dt} \propto -\Theta$ ↗ ↑-01 1 [correct representation of proportionally]

$\frac{d\Theta}{dt} = -k\Theta$ 1 [Formation of D.E]

$\int \frac{d\Theta}{\Theta} = \int -k\,dt$ 1 [Recognises Integration]

$\ln\Theta = -kt^2 + c$ 1 [correct Integrates]

At $t = 0$, $\Theta = 1000$ 1 [Identifies any conditions]

$\ln 1000 = -k(0) + c$

$c = \ln 1000$ 1 [correct output]

After $t = 1$, $\Theta = 1000 - 20 = 980$ ↙ m-01 1 [Correct amount remaining in the tank]

$\ln 980 = -k(1) + \ln 1000$

$k = \ln 1000 - \ln 980$

$k = \ln\left(\frac{1000}{980}\right)$ 1 [correct output]

$\ln\Theta = -t\ln(1.0204) + \ln 1000$

Θ

Amount remaining after 3 hours

$\ln \theta = -3 \ln(1.0204) + \ln 1000$ 1 [Correct substitution]

$\ln \theta = \ln((1.0204)^{-3} 1000)$

$\theta = 941.214$ 1 [Correct Output]

Amount remaining is 941.214 litres

Since the amount remaining is less than 950 litres, the farmer should reduce the opening of the tank tap.

1 [Decision]

TT : 19 scores

2/10

ITEM 3

Obtaining shelf life

$3^{2x} - 4(3^x) = -3$

Let 3^x be y

$y^2 - 4y + 3 = 0$ 1 [Formation a quadratic Equation] T-01

$y^2 - 3y - y + 3 = 0$

$y(y-3) - (y-3) = 0$ 1 [Correctly Solves the quadratic Equation] m-01

$(y-1)(y-3) = 0$

$y - 1 = 0$, $y - 3 = 0$

$y = 1$ $y = 3$ 1 [outputs] → m-01 (any of 2 values)

For $y = 1$ For $y = 3$

$3^x = 1$ $3^x = 3$

$3^x = 3^0$ $x = 1$ } 1 [out puts] $x = 0$ and $x = 1$ } m-01 both Correct

$x = 0$

1 [Forming Index equations]

Shelf life is- - - -which is less than 2 years, therefore the group should not produce- - - - - 1 [Decision] D-01

Increase by constant number each week

Let amount in week 1 be a 1 [Defines Variables

Let additional amount per week be d 1 [Identifies Arithmetic progress]

Amount in week 2 = a + d

Amount in week 3 = a + d + d T-01 for a = d =

This is an arithmetic progression with intial amount as , a and common difference, d

Amount in n^{th} week $L_n = a + (n-1)d$

In 7th week; $L_7 = a + (7-1)d$

$27 = a + 6d$ 1 [Formations Any Simultenous Equations]

m-01

In 12th week; $L_{12} = a + (12-1)d$

$47 = a + 11d$

$a + 6d = 27$

$a + 11d = 47$

$5d = 20$ 1 [Solves simultaneous Equations]

$d = 4$

$a + (6 \times 4) = 27$ } 1 [Outputs]

$a = 3$

$d = 4$ and $a = 3$ } m-01 both correct

Amount in 36th week

$L_{36} = 3 + (36-1)4$ → m-01

$= 143$ 1 out put

Total products upto 36th week

$S_n = \frac{n}{2}(a + L_n)$ m-01

$S_{36} = \frac{36}{2}(3 + 143)$ 1 [Correct substitutions in Correct Formula]

$= 2628$ m-01 1 [Correct output]

Total number of products required upto 36th week is 2628 pieces.

Since it is less than 2630, the group will not recruit additional workers 1 [Decision]

Since, $2628 < 2630$ D-01

Formation of 3 member committee

Options

1 Boy and 2 girls

2 boys and 1 girl

Number of ways T-01

m-01 ${}^3C_1 \times {}^2C_2$ $3 \times 1 = 3$ } 1 [Correctly identifies Combination]

${}^3C_2 \times {}^2C_1$ $3 \times 2 = 6$ 1 [Correct output]

=9 ← Total number of ways $3 + 6 = 9$ 1 [Identifies Addition]

1 [Correct output]

Since the number of possible selection exceeds 5, the school administration should appoint the committee members 1 ✓

TT : 29 Score: [Decision]

Since $9 > 5$ D-01

ITEM 4

T-01 for x, y

Let the numbers of the solar panels be x and y 1 [variables]

M-01
$xy = 72$ 1 [Forms Equation]
$xfy = 18$ 1 [Forms Equation]
$y = 18 - x$ Formulation of correct Equation

m-01 $x(18-x) = 72$

$x(18-x) = 72$ 1 [correct substitution in correct Equation]

Rmd -01 $18x - x^2 - 72 = 0$

$x^2 - 18x + 72 = 0$ 1 [Formation of Quadratic Equation]

$x^2 - 12x - 6x + 72 = 0$

m-01 $(x-6)(x-12) = 0$ 1 [Solves quadratic Equation]

$x - 6 = 0 \quad x - 12 = 0$

m-01 $x = 6 \quad x = 12$ 1 [correct outputs]

$y = 18 - 6 \quad y = 18 - 12$

m-01 $y = 12 \quad y = 6$ 1 [correct outputs]

The numbers of the solar panels are 6 and 12.

The project should be considered possible since the numbers of the solar panals are atleast 6 and not more than 15. 1 [Coirect decission] Decision D-01

Addition of water storage tanks.

1st year	2nd	3rd
4	8	16

T-01

Common ratio $\frac{8}{4} = 2$ and $\frac{16}{8} = 2$ 1 [Identifies G.P]

1 [output r]

This is a geometric progression with common ratio,

$r = 2$, $a = 4$

Cummolative number of tanks

$S_n = a\left(\frac{r^n - 1}{r-1}\right)$ m-01

$508 = 4\left(\frac{2^n - 1}{2-1}\right)$ 1 [Correct substitution in correct Formular]

$127 = 2^n - 1$

$128 = 2^n$ [$\Rightarrow 2^7 = 2^n$ 1 [Identifies Indices]

n=7 m-01 $n = 7$ 1 [Output] 1 [Factorisation]

Comment $n = 7$ or 5 D-01

The required for the cumulative trains to reach 50% is 7 years since it's exceeds 5 years, the council will not provide funding for the upgrading of the existing water supply system.

$$ {}^{n}P_{r} = \frac{n!}{(n-r)!} \quad | \text{ [Identifies Permutation]} $$

$n = 6, \; r = 4$ | [correct substitution]

$$ {}^{6}P_{4} = \frac{6!}{(6-4)!} \quad | \text{ [manipulates formula Notation]} $$

T-01

$$ = \frac{6!}{2!} = \frac{6 \times 6 \times 4 \times 3 \times 2 \times 1}{2 \times 1} $$

m-01

$$ = 360 \quad | \text{ [output]} $$

[Computation Simulation should ~~not~~ be] [Decision used.

DECISION D-01

TT: 20 Scores

x/15

Item 5

T-01 for knowing X and correctly Identifies

Sale (x)	No of students (f)	fx	fx^2
1	2	2	2
2	8	16	32
3	12	36	108
4	5	20	80
5	3	15	75
	$\Sigma f = 30$	$\Sigma fx = 89$	$\Sigma fx^2 = 297$

1 [For x^2 or fx^2]

1 [Correct. fx]

1 [Output correct Σfx]

Variance $= \frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2$ 1 [Correct fx^2]

$= \frac{297}{30} - \left(\frac{89}{30}\right)^2$ 1 [Correct substitution in the formular]

$= 1.0989$ m-01 1 [output]

Comment 1.0989 > 1 D-01

Since variance is greater than 1, the administration should direct the committee to stop selling the new item. 1 [Decision]

Expected number of meals had

$E(x) = \Sigma x P(x = x)$ 1 [Identifies expectation]

$= (1 \times 0.16) + (2 \times 0.35) + (3 \times 0.41) + (4 \times 0.08)$ 1 [Correct substitution]

$= 3.76$ 1 [Output] m-01

2.4 ✓

Since the expected number of meals had is greater than three, the Canteen can continue opening three times a day. 1 [Decision]

Decision / D-01

Bihomial

$X \sim B(20, 0.25)$ T-01

$P(X \geq 8) = 0.1018$ m-01

Comment - $0.1018 < 0.5$ D-01

Item 5

Comparison of sales at the Canteen

Quarterly basis — 4 point moving averages.

Implied moving totals

Moving averages

MA ⇒ 64·5, 65·5, 66, 66, 66·5 All m-01

$\frac{60+84+42+72}{4} = 64{\cdot}5$

$\frac{84+42+72+64}{4} = 65{\cdot}5$

1 [All correct Moving Totals]

$\frac{42+72+64+86}{4} = 66$

1 [All correct MA].

$\frac{72+64+86+42}{4} = 66$

4,5

$\frac{64+86+42+74}{4} = 66{\cdot}5$

$\frac{86+42+74+67}{4} = 67{\cdot}25$

$\frac{42+74+67+x}{4} = 67$ 1 [Reading value from graph]

1 [Forming Equation]

$x = 85$ 1 [Output]

Estimated sales in the 10th term is Ugx 85 million.

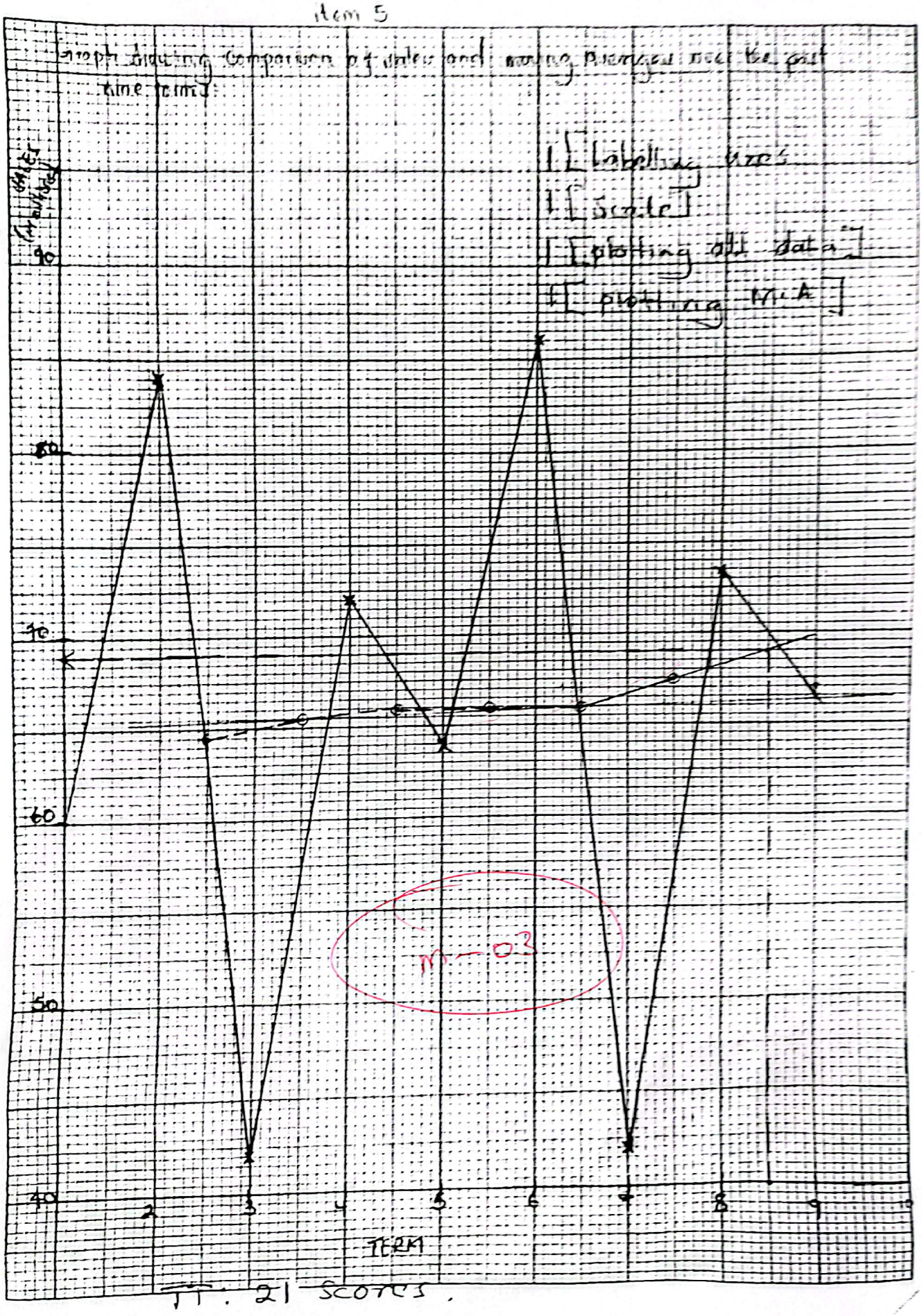
The school administration should consider renewing the contract because estimated sales is greater than Ugx 70 million. 1 [Decision]

$65 \leq MA \leq 69$ — m-01

$77 \leq x \leq 93$ — m-01

Comment: Decision D-01

Item 5



TT. 21 SCORES.

Item 6 T-01 (Number of bags)

Mass (kg)	No of bags (f)	x T-01	fx	fx^2
10-14	4	12	48	576
15-19	6	17	102	1734
20-24	10	22	220	4840
25-29	12	27	324	8748
30-34	8	32	256	8192
	$\Sigma f = 40$		$\Sigma fx = 950$	$\Sigma fx^2 = 24,090$

1 [Identifies fx^2 or x^2]

1 [Σfx^2]

Any of Σfx or Σfx^2 m-01

$$\text{Variance} = \frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2$$

$$= \frac{24090}{40} - \left(\frac{950}{40}\right)^2$$ 1 [Correctly substitutes in correct formula]

$= 38.1875 \text{ kg}^2$ 1 [Correct Output] m-01 if kg^2 is not there not score

Since the variance exceeds 30, the variation in production is not manageable 1 [Decision] D-01

Let the event that a farmer use improved seeds be S.
Let the event that a farmer use irrigation be I

$P(S) = 0.7$ $P(I) = 0.5$

P(both improved seeds and irrigation) $= P(S \cap I)$.

$P(S \cap I) = P(S) \times P(I)$. 1 [Identifies Independent]

$= 0.7 \times 0.5$ 1 [Substitution]

$= 0.35$ 1 [Output] 0.35 m-01

The agricultural officer should encourage the farmers to use both since the probability is less than 0.4 1 [Decision] D-01

X/15

Expected daily amount of coffee processed.

To get k; (K)

$\int_0^4 f(x)dx = 1$ | [identifies Integration] T-01

$\int_0^4 kx^2 dx = 1$

$\left[\frac{k}{3}x^3\right]_0^4 = 1$ | [Integrates correctly]

| [Probability theory = 1]

$k\left[\frac{4}{3} - \frac{0^3}{3}\right) = 1$ [r | [Correct substitution]

$k = \frac{3}{64}$ | [Correct output] M-01

Expectation:

$E(x) = \int_0^4 x f(x)\,dx$ | [identifies E(x).

$= \int_0^4 x \cdot \frac{3}{64} x^2 dx$ M-01

$= \left[\frac{3}{64} \cdot \frac{x^4}{4}\right]_0^4$ | [Correct integration and substitution of limits]

$= \frac{3}{64}\left(\frac{4^4}{4} - \frac{0^4}{4}\right)$

= 3 tonnes. M-01! output; even when the units are not there

The agricultural officer should encourage the expansion of

Item 6.

labelling the axes correctly 1 – 0

A graph showing the amount of fertilizer used and the corresponding coffee yield obtained.

Correct
- Labelling and Scale
- Plotting all data
- Plotting and drawing line of best fit

(1 mark)

(2 marks)

Estimated coffee yield when 9kg of fertilizer is used is 27 bags

(1 mark) for correct estimate

Since it does not exceed 30 bags, the agricultural officer should encourage appropriate fertilizer use.

– No starting values do not award if done.

fertilizer used (kg)

Scores.